

SIH26103 — PAIMANA AI INTELLIGENCE LAYER

Final Team Concept & End-to-End Product Definition

THE CORE IDEA

We are not building a replacement for PAIMANA. We are proposing an AI-powered predictive and early-warning intelligence layer around the existing PAIMANA ecosystem. It uses historical OCMS + PAIMANA project data and current project information to predict cost overruns, time overruns and implementation risks, explain the reasons, benchmark projects, generate early warnings and help MoSPI prioritise intervention.

1. The SIH Problem Statement — What We Must Solve

- MoSPI's Infrastructure & Project Monitoring Division (IPMD) monitors Central Sector Infrastructure Projects costing ₹150 crore and above across infrastructure Ministries/Departments.
- OCMS was used for project monitoring since 2006 and created a valuable historical repository containing project cost, expenditure, timelines, implementation status, cost overruns and time overruns.
- OCMS was later modernized into PAIMANA (Project Assessment, Infrastructure Monitoring and Analytics for Nation-building), a web-based integrated project-monitoring ecosystem.
- PAIMANA captures approved cost, revised cost, expenditure, implementation timelines, physical progress, milestones, implementing agencies and project status. Project information is updated monthly through role-based access and APIs.
- The SIH opportunity is to move from descriptive monitoring to predictive and prescriptive monitoring using AI/ML, statistical analysis, forecasting, big-data analytics and/or LLMs.

2. The Problem in Simple Words

PAIMANA already stores and monitors a large amount of project information. The missing intelligence is: What is likely to happen next, why is that risk increasing, and which project should the monitoring team look at first?

- Reported progress can lag behind what is happening on the ground.
- Delays may become obvious only after milestones are missed.
- Cost overruns are generally visible as an outcome; the goal is to forecast escalation earlier.
- Thousands of projects cannot be manually prioritised with equal attention.
- A dashboard can show that a project is delayed without clearly explaining the underlying drivers.
- Comparing very different projects without suitable peer groups can produce misleading conclusions.

3. What Exactly Are We Building?

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Our product is an **AI/ML risk-intelligence layer on top of PAIMANA**. It turns project data into forward-looking decision support.

- **Predict:** estimate the probability of cost overrun and time overrun from historical patterns and current project conditions.
- **Score:** produce project-level cost risk, time risk, implementation risk and an overall risk score.
- **Explain:** show the major factors contributing to a project's risk in understandable language.
- **Benchmark:** compare a project with appropriate similar projects rather than judging it in isolation.
- **Warn:** identify rising risk before a major delay or cost escalation becomes official.
- **Prioritise:** help monitoring teams identify which projects need attention first.
- **Assist:** provide an LLM-enabled project intelligence assistant for natural-language questions.
- **Verify:** where additional execution evidence is available, flag inconsistencies for human review rather than automatically declaring wrongdoing.

4. Complete Concept Flow



5. PAIMANA Foundation vs Our Proposed Additional Evidence

This distinction is critical: **PAIMANA is the existing foundation**. Our additional site/execution information is a proposed enhancement that should be tested to determine whether it improves prediction and early-warning capability.

Existing PAIMANA / CUF foundation	Proposed additional execution-level information
Approved and revised cost	More frequent project/site records where feasible
Expenditure to date	Site photographs and supporting evidence
Physical progress %	Work/activity and relevant quantity records
Milestones and timelines	Material receipt/use information where relevant
Implementing agency	Structured delay and bottleneck reasons
Project status and other available fields	Inspection/visit trail and verification records
Other CUF fields available in the actual data	Bill/document vs reported progress inconsistency signals

6. Ground-Research Differentiator

Our construction-side understanding can be used as a **research input**: identify execution factors that may matter in real projects but may not be represented in existing CUF fields. The important point is to **test** whether these factors improve prediction, not assume that they do.

- **Payment-delay → cash-flow → manpower chain**: delayed payments can affect contractor cash flow and subsequently labour/material mobilisation.
- **Subcontracting structure**: the relationship between a main contractor and subcontractors/petty contractors can affect execution coordination and resource availability.
- **Utility shifting / land clearance**: unresolved site access, utility relocation or land-related constraints can create schedule friction.
- **Design/scope changes**: repeated changes can affect planning, procurement, cost and schedule.
- **Billing-vs-physical-progress mismatch**: unusual differences between reported billing/expenditure and physical progress can become a verification signal.

Research experiment: evaluate a baseline model using existing CUF/PAIMANA variables, then evaluate an expanded model with validated additional variables. Report whether the added information improves predictive performance and/or warning lead time.

7. Daily Project / Site Record

The original daily-record idea is useful as a **proposed evidence layer**. It should not be presented as an existing PAIMANA feature unless the actual PAIMANA specification confirms it.

- Create a chronological project timeline from execution start to completion.
- For relevant days/activities, record site status, work completed, resources/issues and supporting evidence.
- Attach dated photographs to a project/activity record when appropriate.
- Capture material unloading/receipt and supporting documents where relevant.
- Record labour, resource and machinery constraints when they materially affect execution.
- If work is stopped or delayed, capture a structured reason and optional explanation.

- Maintain project/date/responsible-role/evidence/follow-up information so the record remains traceable.

8. Delay Intelligence Workflow

- Receive the latest project status.
- If work is progressing, continue monitoring.
- If work is stopped or materially affected, capture the reason.
- Possible categories: labour unavailable, material/resource unavailable, machinery/tool unavailable, land/legal/utility constraints, design/scope issue, contractor/subcontractor issue, or other bottleneck.
- Assign the issue to the relevant responsible team/role for explanation or follow-up.
- Repeated unresolved issues can contribute to the implementation-risk picture and be surfaced to the monitoring administrator.

9. Cost Monitoring & Bill Verification

The earlier idea about inflated bills should be framed as an **anomaly/inconsistency detection and verification workflow**, not a corruption-detection engine.

- Link reported expenditure/bills with available project context and progress records.
- Compare claimed progress/quantities with available evidence and historical/project patterns.
- If an unusual mismatch is detected, generate a review flag.
- Ask the authorised responsible team for clarification or supporting evidence.
- Record who submitted, who reviewed, what evidence was considered and what follow-up occurred.
- AI identifies anomalies; authorised officials perform verification and make administrative decisions.

10. Officer Inspection & Accountability Trail

- Record project/inspection visits with date, responsible person and supporting evidence where the workflow permits.
- Maintain a traceable history of reported issues and follow-up actions.
- If a required inspection/follow-up is overdue, the platform can generate a reminder or escalation signal for the responsible administrator.
- The objective is accountability and traceability.

11. Historical Data → Prediction: The Main AI Value

The strongest part of the SIH problem statement is the historical database created through OCMS and PAIMANA. Instead of using old data only for charts and reports, we use it to learn patterns that can support prediction.

- **Historical:** study completed/ongoing project journeys and their outcomes.
- **Learn:** identify relationships between project variables and later cost/time overruns or implementation risks.
- **Current:** analyse the latest available data for an ongoing project.
- **Predict:** estimate the probability of the target risk.
- **Explain:** identify the strongest contributing factors.
- **Warn:** surface a warning when risk becomes materially elevated or the trend deteriorates.
- **Intervene:** the authorised monitoring team decides what action is appropriate.

Illustrative example: If historical projects show that low progress relative to plan, high expenditure relative to progress and repeated milestone slippage are associated with later schedule overruns, a current project showing a similar pattern may receive a high time-overrun probability. This is a probability-based warning, not a guarantee.

12. Risk Scoring Framework

Risk	Meaning	Officer view
Cost Risk	Likelihood of future cost escalation	Probability, trend and major drivers
Time Risk	Likelihood of schedule/time overrun	Probability, milestone pressure and drivers
Implementation Risk	Emerging execution/bottleneck risk	Issues, patterns and evidence
Overall Risk	Priority signal across dimensions	Which projects need attention first

13. Benchmarking & Comparative Analytics

- Compare a project with relevant peers: sector, project type, scale, implementation stage and other suitable attributes.
- Highlight projects that are unusually slow, unusually high-spending relative to progress, or unusually risky compared with comparable projects.
- Avoid treating all projects as directly comparable.
- Use benchmarking to support prioritisation and learning, not to automatically assign blame.

14. Cost Escalation Driver Analysis

The system should answer not only “**What is the risk?**” but also “**Why is the model giving this risk signal?**”. A driver-analysis module can show the most influential variables for an individual prediction.

- Show project-level contributing factors in plain language.
- Use explainability/feature-importance methods to support the explanation.
- Distinguish predictive contribution from proof of causation.
- Allow authorised users to inspect underlying project records where available.

15. LLM-Enabled Project Intelligence Assistant

- An authorised officer can ask: “Why is Project X high risk?”
- Summarise current project status, risk dimensions and major contributing factors.
- Explain recent milestone slippage or expenditure/progress patterns.
- Surface relevant project records/evidence available to the user.
- Clearly distinguish recorded facts from model-generated interpretation.

16. Portfolio-Level Management View

- See monitored projects in one prioritised view.
- Filter by risk, sector, ministry/department, project stage and other permitted fields.
- Identify high-risk projects and projects whose risk is increasing.
- Open a project to see risk trend, drivers, benchmark comparison, recent issues and follow-up status.
- Help monitoring teams allocate attention where it can have the greatest value.

17. Direct Mapping to MoSPI's Indicative Expected Outcomes

PS Outcome	Our corresponding module
a. Cost Overrun Prediction Model	Predict cost-escalation probability.
b. Time Overrun Prediction Model	Predict schedule/time-overrun probability.
c. Project Risk Scoring Framework	Provide cost, time, implementation and overall risk views.
d. Early Warning Alert System	Flag emerging risk before a major problem is officially recorded.
e. Benchmarking & Comparative Analytics	Compare projects with appropriate peer groups.
f. Cost Escalation Driver Analysis	Explain factors contributing to cost-risk signals.
g. AI-powered Monitoring Dashboard	Present portfolio/project risk, trends, drivers, alerts and comparisons.
h. LLM-enabled Project Intelligence Assistant	Allow natural-language project questions and explanations.
i. Documentation & Deployment Framework	Document data, assumptions, evaluation, limitations and deployment.

18. Phase 1 vs Future Roadmap

For the hackathon prototype, separate what can be credibly demonstrated with available data from what we want to show as a future extension.

Phase 1 — Core prototype / demo	Phase 2 — Future extension / roadmap
CUF/PAIMANA-based cost & time prediction	Daily geo-tagged photo progress verification
Risk scoring + risk trend + early warning	Automated billing-vs-progress anomaly workflow
Benchmarking	Officer visit/reminder/escalation workflow
Cost driver/explainability analysis	Full role-based site-to-MoSPI execution workflow
LLM project intelligence assistant	Richer live site evidence and additional ground-level variables
AI vs conventional statistical model comparison	Broader operational integrations, subject to access/data availability
CUF-only vs CUF + validated additional-factor experiment	Production-grade verification and governance workflows

Why this split matters: the official PAIMANA data described in the PS is updated monthly and is fundamentally project-monitoring data. A full live daily-photo/field-verification system should therefore be presented as an extension unless the required operational data and access are available.

19. Who Controls the Solution?

Role	Purpose
MoSPI / IPMD	Monitoring authority and primary decision-support user.
Project Monitoring Administrator	Operational control: review warnings, prioritise cases, assign follow-ups and monitor closure.
Implementing Agency / Project Team	Provide updates, explanations and supporting project information as permitted.
Site / Engineering Roles	Provide execution-level records/evidence where the proposed workflow includes them.

20. What We Must NOT Claim

- We are not replacing PAIMANA; we are adding predictive intelligence around it.
- AI cannot prove corruption. It can identify anomalies or inconsistencies that deserve human verification.
- AI cannot know the future with certainty. It estimates probabilities based on learned patterns and available data.
- A photograph alone should not be presented as proof of exact construction quantity; evidence needs project context and appropriate records.
- AI does not replace engineers, officers or authorised decision-makers.
- More variables do not automatically mean a better model; their predictive value must be evaluated.

21. Strongest One-Line Differentiator

“Use the historical intelligence already present in OCMS + PAIMANA to predict risk in today's projects, then test whether real-world execution factors missing from CUF can make those predictions earlier and better.”

22. Team Explanation — Short Version

We are building an AI-powered predictive intelligence layer on top of MoSPI's existing PAIMANA project-monitoring ecosystem. PAIMANA already contains current project-monitoring information, while OCMS and PAIMANA together provide nearly two decades of historical project data. We will learn from that historical data to identify patterns associated with cost overruns, time overruns and implementation risks. When current project data is analysed, the system will estimate the probability of these risks, show the major drivers, compare the project with similar projects, track the risk trend and generate early warnings so MoSPI can prioritise intervention before a problem becomes a major overrun. In addition, we propose testing selected ground-level execution factors—such as payment-delay/cash-flow effects, subcontracting structure, land or utility constraints, design changes, structured delay reasons and evidence-based progress/billing inconsistencies—to determine whether they improve prediction beyond existing CUF fields. The solution therefore moves PAIMANA from mainly descriptive monitoring toward predictive, explainable and actionable decision support.

23. Final Product Definition

PAIMANA / OCMS HISTORY → Current PAIMANA Data → Validated Ground-Level Variables → AI/ML Prediction → Risk Score + Explanation → Benchmarking → Early Warning → Prioritised Human Intervention

This document defines the product concept and scope. Detailed technical implementation—dataset schema, exact CUF fields, target definitions, model selection, evaluation metrics, APIs, UI screens, permissions and deployment—comes after the team agrees on the product.